Practice workbook · Turn a flowchart or pseudocode into Python

The algorithm is already designed - as numbered pseudocode, a flowchart or a trace table plus requirements - and the student implements its logic in Python inside a given structure (5-11 marks).

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-CODE-04

1. Question 3 · part c · 10 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part c only.

3 Algorithms can be represented in flowcharts, pseudocode or program code.

(a) Trace tables can be used with flowcharts or pseudocode.

Give **two** characteristics of a trace table.

(2)

1 .....

.....

2 .....

.....

(b) An algorithm has been written to validate numbers entered by the user.

Figure 2 shows the pseudocode for the algorithm.

```
1  RECEIVE num1 FROM (INTEGER) KEYBOARD
2  RECEIVE num2 FROM (INTEGER) KEYBOARD
3  IF (((num1 < 16) AND (num2 < 23)) OR (num2 = 13)) THEN
4      SEND "State 1" TO DISPLAY
5  ELSE
6      IF ((num2 > 12) AND (num1 > 20)) THEN
7          SEND "State 2" TO DISPLAY
8      ELSE
9          IF ((num1 = 88) OR NOT (num2 = 18)) THEN
10             SEND "State 3" TO DISPLAY
11         ELSE
12             SEND "State 4" TO DISPLAY
13         END IF
14     END IF
15 END IF
```

Figure 2

Complete the table to show the output for each set of inputs.

(3)

num1	num2	Output
88	18	
17	18	
12	19	

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P 7 2 9 3 8 A 0 6 1 6

1. Question 3 · part c · 10 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part c only.

(c) A program is required to calculate the result of raising one integer (the base) to the power of another (the exponent).

Figure 3 shows a flowchart for the algorithm.

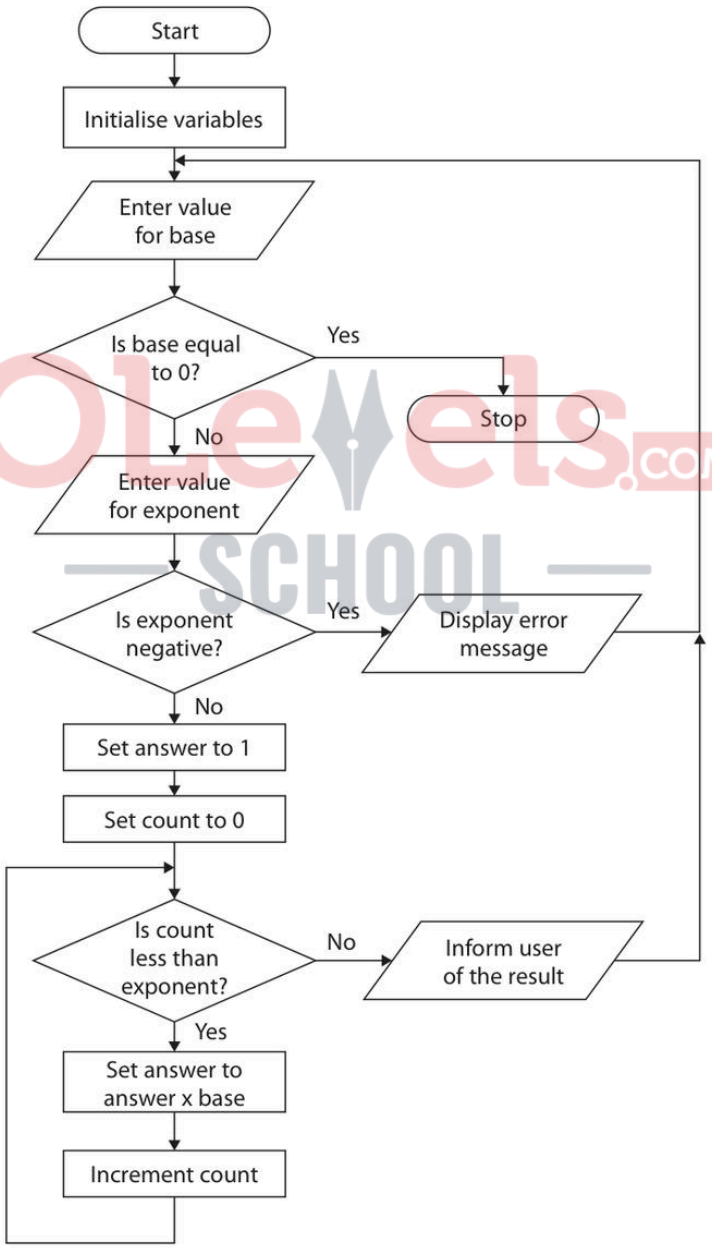


Figure 3

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1. Question 3 · part c · 10 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part c only.

The program has these requirements:

- outputs meaningful error messages
- outputs the final answer with the base and the exponent.

Open **Q03c** in the code editor.

Write a program to implement the logic in the flowchart.

Do not add any further functionality.

Save your code as **Q03cFINISHED** with the correct file extension for the programming language.

(10)

(Total for Question 3 = 15 marks)

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1. Question 3 · part c · 10 marks · Working and answer

Working and answer

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2. Question 2 · part a · 10 marks

2019 JUN · Paper 02

Further practice · Attempt part a only.

2 Anna has a 5-year-old daughter, Beatrice, and a 14-year-old son, Graham.

(a) Anna wants to program a guessing game for Beatrice to play on the computer.

The program will generate a random number between 1 and 10.

Beatrice has to guess the number.

This pseudocode contains the logic required to create the game.

```
1   #   Initialise variables
2       SET counter TO 1
3       SET answer TO RANDOM(9) + 1     # i.e. a random integer 1 to 10 inclusive
4       SET guess TO 0
5
6   #   Print prompt and take guess from user
7       SEND "Enter a number from 1 to 10: " TO DISPLAY
8       RECEIVE guess FROM (INTEGER) KEYBOARD
9
10  #   Create while loop to check guess
11      WHILE guess <> answer DO
12          SET counter TO counter + 1
13          IF guess > answer THEN
14              SEND (guess & " was too high. Try again.") TO DISPLAY
15          ELSE
16              SEND (guess & " was too low. Try again.") TO DISPLAY
17          END IF
18          SEND "Guess again: " TO DISPLAY
19          RECEIVE guess FROM (INTEGER) KEYBOARD
20      END WHILE
21
22  #   Report the correct answer to the user and display the number of guesses
23      SEND ("You guessed " & guess & " in " & counter & "guesses.") TO DISPLAY
```

Write a program to implement the logic in the pseudocode.

Open **Q02a** in the code editor.

You **must** use the structure give in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

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P 6 1 8 8 1 R A 0 5 1 6

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Turn over ▶

2. Question 2 · part a · 10 marks

2019 JUN · Paper 02

Further practice · Attempt part a only.

(b) Anna has written a program to help Graham with calculations for his science lessons.

Open **Q02b** in the code editor.

Use the code to answer these questions.

(i) State the value of **distance** when the program finishes.

(1)

(ii) Give **one** reason why there are brackets around **newSpeed + initialSpeed**

(1)

(iii) State the data type for the variable **distance**.

Justify your answer.

(2)

Data type

Justification

(iv) State the value of **distance** when **acceleration** is 2.75

(1)

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P 6 1 8 8 1 R A 0 6 1 6

2. Question 2 · part a · 10 marks · Working and answer

Working and answer

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3. Question 2 · part a · 10 marks

2020 NOV · Paper 02

Further practice · Attempt part a only.

2 Rama wants some computer programs for his daughter, Ayomi.

(a) Rama wants a program to calculate the area of a triangle.

(area of a triangle = ½ base x height)

This pseudocode contains the logic required to create the program.

```
1 #  Initialise variables
2
3 SET base TO 50
4 SET heightChk TO TRUE
5
6 #  Print prompt and take input from user
7
8 WHILE heightChk DO
9     SEND "Enter the height (between 1 and 100): " TO DISPLAY
10    RECEIVE height FROM (INTEGER) KEYBOARD
11    IF (height >= 1 AND height <= 100)
12    THEN heightChk = FALSE
13    END IF
14 END WHILE
15
16 #  Calculate and print out values
17
18 SET area TO 0.5 * base * height
19
20 SEND ("Base of triangle is: " , base) TO DISPLAY
21 SEND ("Height of triangle is: " , height) TO DISPLAY
22 SEND ("Area of triangle is: " , area) TO DISPLAY
```

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3. Question 2 · part a · 10 marks

2020 NOV · Paper 02

Further practice · Attempt part a only.

Write a program to implement the logic in the pseudocode.

Open **Q02a** in the code editor.

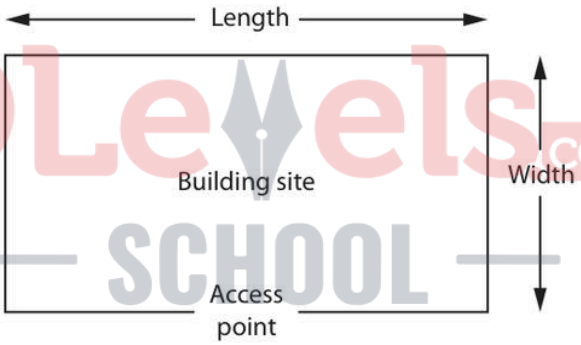
You must use the structure given in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

(b) Ayomi wants a program that will calculate the number of fence panels required to surround building sites such as the one shown in the diagram (not to scale).



Each building site is a rectangle.

Each fence panel is 1 metre long.

A gap of 4 metres is needed to access each site. The location of the gap is not significant.

The user will input the length and width of the building site as whole numbers of metres.

The program will output the minimum number of full panels needed.

Q02b provides a structure for the program.

Open **Q02b** in the code editor.

Write the program.

You must use the structure given in **Q02b** to write the program.

Do not add any further functionality.

Save your code as **Q02bFINISHED** with the correct file extension for the programming language.

(7)

(Total for Question 2 = 17 marks)

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3. Question 2 · part a · 10 marks · Working and answer

Working and answer

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4. Question 2 · part a · 11 marks

2021 JUN · Paper 02

Further practice · Attempt part a only.

2 A book club uses computer applications.

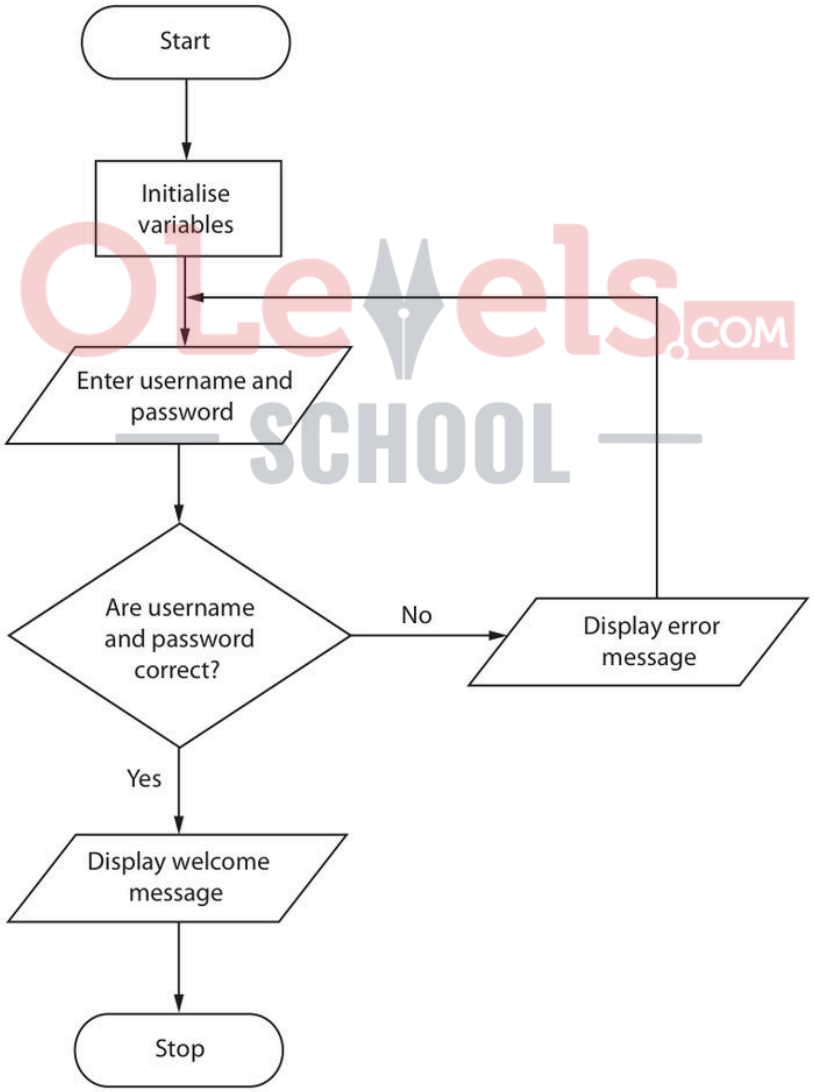
(a) The club wants a program to ensure that logins are valid.

A flowchart for a test version of the program has been designed.

The test version uses:

- a username of bard423
- a password of nX2934?

Here is the flowchart.



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4. Question 2 · part a · 11 marks

2021 JUN · Paper 02

Further practice · Attempt part a only.

Write a program to implement the logic in the flowchart.

Open **Q02a** in the code editor.

You must use the structure given in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(11)

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4. Question 2 · part a · 11 marks

2021 JUN · Paper 02

Further practice · Attempt part a only.

(b) The club sells books.

It needs a computer program to monitor:

- the number of books sold
- the amount of profit made.

Q02b provides a structure for the program.

Open **Q02b** in the code editor.

Amend the code to complete the If statement used to produce the outputs described in the table.

Condition	Output message
Number of books sold is at least 5 and profit made is at least 10	Sales and profit are good this week
Number of books sold is over 20 and profit made is at least 20	Sales and profit are excellent this week
Number of books sold is under 5 or profit made is under 5	Poor performance this week
All other inputs	Alert manager

Do not add any further functionality.

Save your code as **Q02bFINISHED** with the correct file extension for the programming language.

(4)

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4. Question 2 · part a · 11 marks · Working and answer

Working and answer

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5. Question 2 · part a · 10 marks

2021 NOV · Paper 02

Further practice · Attempt part a only.

2 Estelle is learning about sorting and searching algorithms.

(a) Estelle's friend has given her a partially completed program to carry out a binary search.

This pseudocode contains the logic required to complete the program.

```
11 RECEIVE item FROM (INTEGER) KEYBOARD
12 WHILE start <= end AND NOT found DO
13      SET middle TO (start + end) DIV 2
14      IF numberList[middle] = item THEN
15       SET found TO TRUE
16      ELSE
17       IF item < numberList[middle] THEN
18        end = middle - 1
19       ELSE
20        start = middle + 1
21       END IF
22      END IF
23      SET count TO count + 1
24 END WHILE
```

Open file **Q02a** in the code editor.

Amend the code to complete the program.

You **must** use the structure given in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

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5. Question 2 · part a · 10 marks

2021 NOV · Paper 02

Further practice · Attempt part a only.

(b) **Figure 3** shows an array of names.

0	1	2	3	4	5	6	7
Yekatrina	Juan	George	Elija	Christine	Sangeetha	Melanie	Shu-fen

Figure 3

Explain why a binary search algorithm would not find 'Juan' in the array.

(4)

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5. Question 2 · part a · 10 marks · Working and answer

Working and answer

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6. Question 2 · part b · 11 marks

2022 JUN · Paper 02

Further practice · Attempt part b only.

2 Raza is writing a program to tell the user whether a number they input is a prime number.

A prime number is a whole number, larger than one, that can only be divided by one and itself with no remainder.

This pseudocode contains the logic required to complete the program.

```
FUNCTION checkPrime(pNumber)
BEGIN FUNCTION
    IF pNumber = 1 THEN
        check = False
    ELSE
        check = True
        FOR count FROM 2 TO pNumber DO
            IF pNumber MOD count = 0 THEN
                check = False
            END IF
        END FOR
    END IF
RETURN check
END FUNCTION

SEND "Enter a number: " TO DISPLAY
RECEIVE number FROM (INTEGER) KEYBOARD
SET result TO checkPrime(number)

IF result = True THEN
    SEND (number & " is a prime number") TO DISPLAY
ELSE
    SEND (number & " is not a prime number") TO DISPLAY
END IF
```

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6. Question 2 · part b · 11 marks

2022 JUN · Paper 02

Further practice · Attempt part b only.

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(a) Answer these questions about the pseudocode.

(i) Identify a line number where **selection** starts.

(1)

(ii) Identify the line number where **iteration** starts.

(1)

(iii) Identify a line number where a **string** and a **number** are printed on the same line.

(1)

(iv) Identify the name of a **local variable**.

(1)

(v) Identify the name of a **parameter**.

(1)

(b) Write a program to implement the logic in the pseudocode.

Open **Q02b** in the code editor.

Write the program.

You must use the structure given in **Q02b** to complete the program.

Do not add any further functionality.

Save your code as **Q02bFINISHED** with the correct file extension for the programming language.

(11)

(Total for Question 2 = 16 marks)

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6. Question 2 · part b · 11 marks · Working and answer

Working and answer

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7. Question 3 · part c · 5 marks

2025 JUN · Paper 02

Further practice · Attempt part c only.

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3 A youth club organises age-specific events for its members.

A program is used to find members of the correct age.

(a) The program first sorts the list of members by age.

(i) A bubble sort algorithm is used to sort the list into ascending order.

The table shows a list of ages that must be sorted.

Complete the table to show how the algorithm sorts the list.

You may not need to use all the rows.

(2)

9	11	12	19	14	18	15	17

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(ii) State the number of swaps made in the first pass.

(1)

(iii) State the **least** number of passes needed to complete the sort algorithm for this list.

(1)

7. Question 3 · part c · 5 marks

2025 JUN · Paper 02

Further practice · Attempt part c only.

(b) The youth club stores a list of sports equipment.

The list contains over 1000 items, sorted in ascending order.

State and justify the **most** appropriate search algorithm to find an item in the list.

(2)

Search algorithm

.....

Justification

.....

.....

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7. Question 3 · part c · 5 marks

2025 JUN · Paper 02

Further practice · Attempt part c only.

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(c) A program is required to calculate and display the average age of members taking part in an activity.

Figure 3 shows a flowchart for the algorithm.

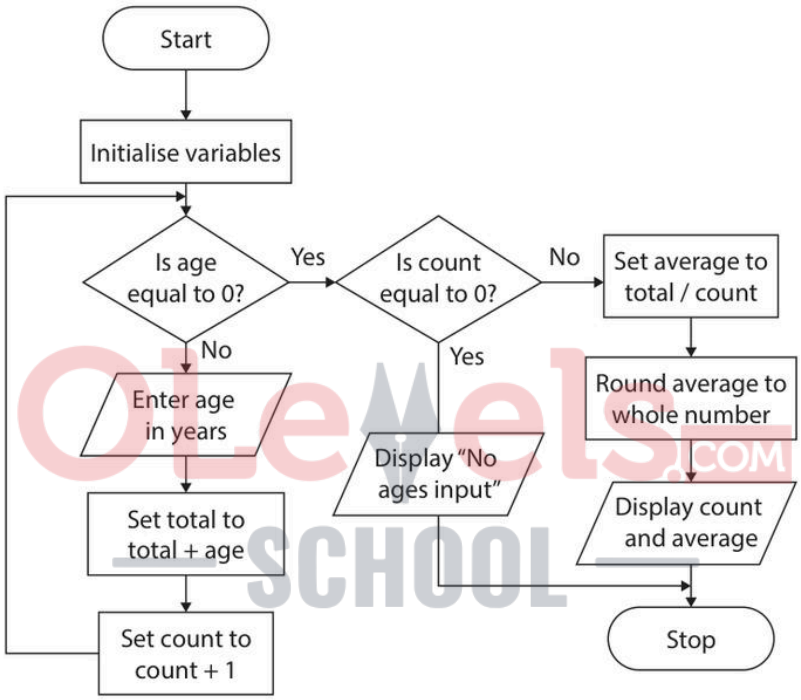


Figure 3

Open **Q03c** in the code editor.

Write a program to implement the logic of the flowchart.

Use the data given in **Figure 4** to test your program.

Test data	Expected output
7 8 9 10 11 0	You entered 5 ages; the average age is 9.
16 16 14 0	You entered 3 ages; the average age is 15.
0	No ages input

Figure 4

Do not add any further functionality.

Save your code as **Q03cFINISHED** with the correct file extension for the programming language.

(5)

(Total for Question 3 = 11 marks)

7. Question 3 · part c · 5 marks · Working and answer

Working and answer

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8. Question 4 · part c · 8 marks

2025 JUN · Paper 02

Further practice · Attempt part c only.

4 Eli is designing a three-throw dice game to play on a computer.

He uses abstraction and decomposition to help him.

(a) **Figure 5** shows how Eli uses comments to plan code before writing it.

Eli's preferred language uses a double-slash // to write comments.

```
1  // roll dice 3 times
2
3     // display number rolled
4
5     // update subtotal
6
7     // check for roll of 3 or 6 and increment booster
8
9  // calculate and output final score
10
```

Figure 5

State why this is a type of decomposition.

(1)

(b) Eli decides to use a library subprogram to generate random numbers that model a dice throw.

Explain why this is an example of abstraction.

(2)

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8. Question 4 · part c · 8 marks

2025 JUN · Paper 02

Further practice · Attempt part c only.

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(c) Eli's program has these requirements.

- A roll of the dice is a random number between 1 and 6.
- The dice is rolled three times.
- The numbers rolled are added to a subtotal.
- The final score is calculated by multiplying the subtotal by a booster value.
- The booster value starts at 1.
- Each time a 3 or 6 is rolled, the booster value is incremented by 1.
- The game outputs the value of each roll, and the final score.

Figure 6 shows a trace table for one execution of the program.

Output	Subtotal	Booster
Rolled: 3	0 + 3 = 3	3 rolled, so incremented to 2
Rolled: 2	3 + 2 = 5	Neither 3 nor 6 rolled
Rolled: 6	5 + 6 = 11	6 rolled, so incremented to 3
Score 33	Subtotal (11) * Booster (3) = 33	

Figure 6

Open **Q04c** in the code editor.

Write the code to create this program.

You must use the layout, variables and constants given in **Q04c**.

Do not add any further functionality.

Save your code as **Q04cFINISHED** with the correct file extension for the programming language.

(8)

(Total for Question 4 = 11 marks)

8. Question 4 · part c · 8 marks · Working and answer

Working and answer

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